



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance

3,900

Purchases

18

Columns

\$50.76

Avg Spend

3.75

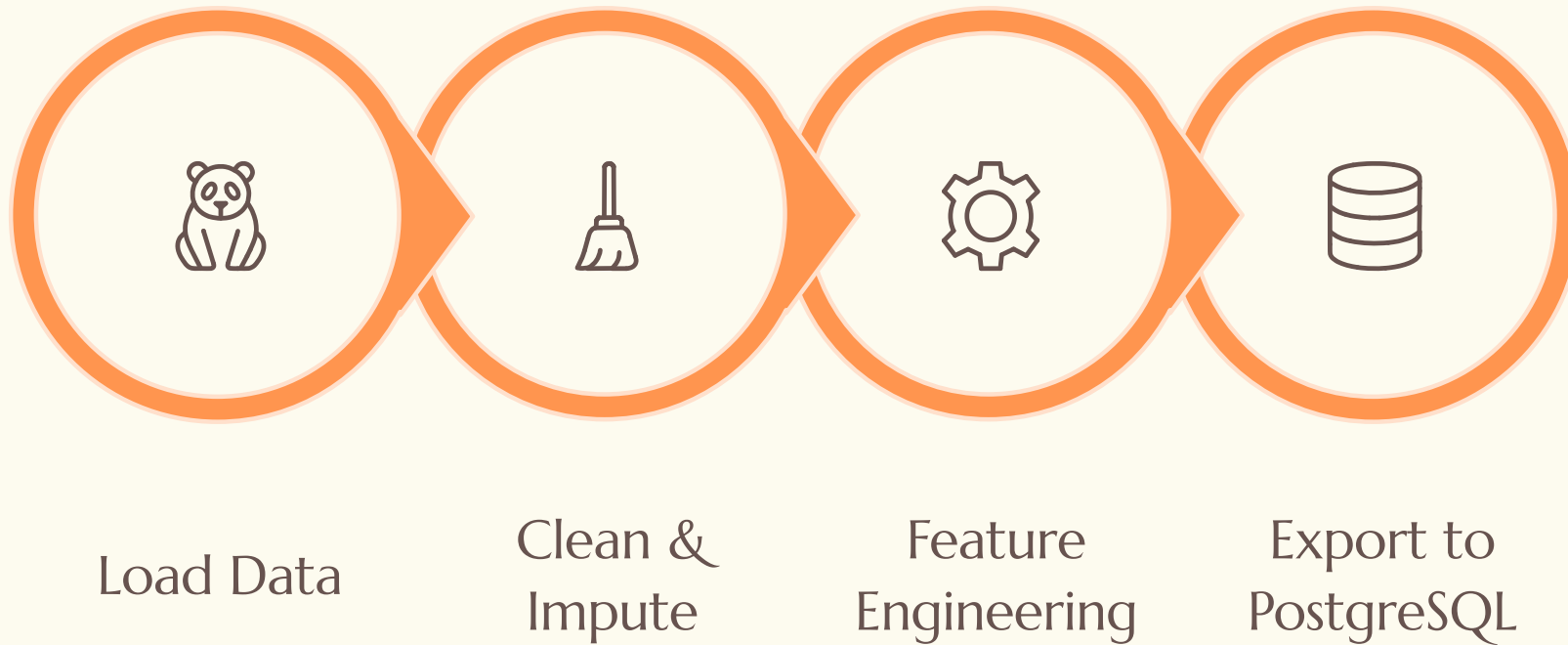
Avg Rating

Key Features

- Demographics: Age, Gender, Location, Subscription
- Purchase: Item, Category, Amount, Season, Size
- Behavior: Discounts, Frequency, Ratings, Shipping

⚠ 37 missing values in Review Rating — imputed using category median

Data Preparation in Python



Columns renamed to snake_case; promo_code_used dropped after confirming redundancy with discount_applied.

Revenue by Gender

Male

\$157,890 total revenue

Female

\$75,191 total revenue

High-Spend Discounters

839 customers used discounts yet spent above average



Top Products & Discount Dependency

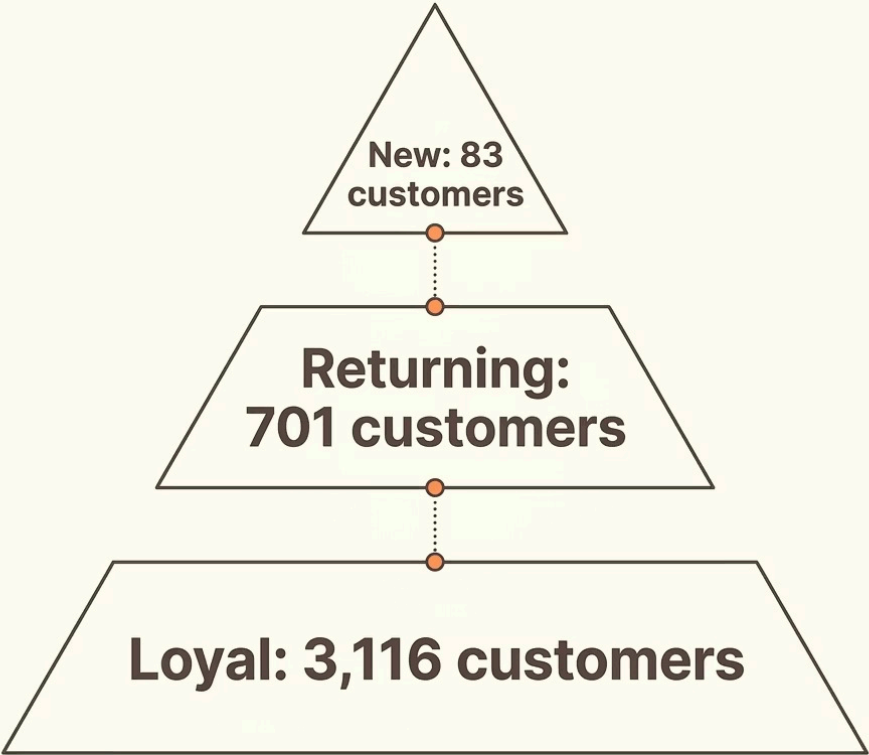
Top 5 by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation

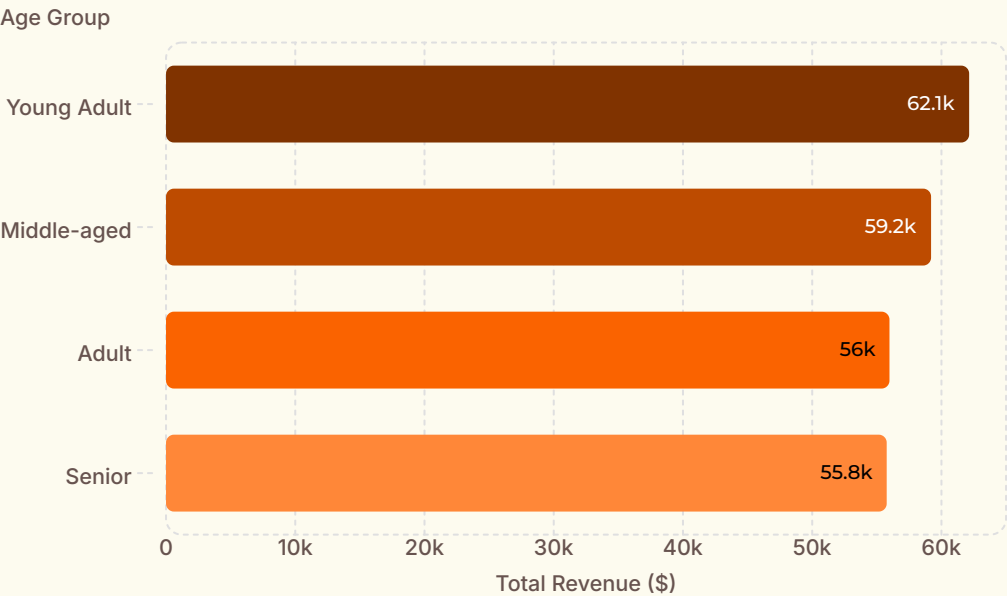


Subscription vs. Non-Subscribers

Status	Customers	Avg Spend	Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

i Repeat buyers (>5 purchases): 958 subscribed vs. 2,518 non-subscribed

Revenue by Age Group



Shipping Impact

Shipping	Avg Purchase
Standard	\$58.46
Express	\$60.48

Express users spend ~3.5% more — a high-value segment.

Top Products by Category

Accessories

1. Jewelry — 171
2. Sunglasses — 161
3. Belt — 161

Clothing

1. Blouse — 171
2. Pants — 171
3. Shirt — 169

Footwear

1. Sandals — 160
2. Shoes — 150
3. Sneakers — 145

Outerwear

1. Jacket — 163
2. Coat — 161



Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27% / 73%

Subscribers vs. Non



Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits

→ Loyalty Programs

Reward repeat buyers to grow the Loyal segment

→ Review Discount Policy

Balance sales boosts with margin control

→ Targeted Marketing

Focus on Young Adults and express-shipping users